

Task 1: Customer Analytics

Chip Category Analysis — Strategic Report for Julia (Category Manager)

Dataset Period	Jul 2018 – Jun 2019 (12 months)
Total Transactions	~264,834 after cleaning
Customer Segments	7 Life Stages × 3 Spending Tiers = 21 Segments
Unique Brands	21 brands (after cleaning)
Pack Sizes	21 unique sizes (70g – 380g)

1. Data Understanding

Two datasets were provided and linked via the loyalty card number (LYLTY_CARD_NBR):

Table 1 — Customer Segments (QVI_purchase_behaviour.csv)

Column	Type	Description
LYLTY_CARD_NBR	ID	Unique customer loyalty card number — join key
LIFESTAGE	Category	Customer life stage (7 values)
PREMIUM_CUSTOMER	Category	Spending tier: Budget / Mainstream / Premium

Table 2 — Transactions (QVI_transaction_data.csv)

Column	Type	Description
DATE	Datetime*	Transaction date (converted from Excel serial format)
STORE_NBR	ID	Store identifier
LYLTY_CARD_NBR	ID	Customer ID — join key to Table 1
TXN_ID	ID	Unique transaction identifier
PROD_NBR	ID	Product identifier
PROD_NAME	Text	Full product name — source for brand & pack size
PROD_QTY	Numeric	Number of packets purchased
TOT_SALES	Numeric	Total dollar value of the transaction

2. Data Cleaning & Preparation

2.1 Date Conversion

The DATE column was stored as an Excel serial integer. It was converted to proper datetime format using pandas:

```
data["DATE"] = pd.to_datetime(data["DATE"], origin="1899-12-30", unit="D")
```

After conversion, the dataset spans exactly **12 months: July 2018 – June 2019**. One date was found missing — Christmas Day (25 December 2018) — confirming stores were closed that day.

2.2 Outlier Removal

Inspection of PROD_QTY revealed 2 transactions where a customer purchased 200 packets in a single visit, generating a TOT_SALES value of \$650 — far outside the normal range (max after removal: \$29.50). These rows were identified as wholesale or erroneous entries and removed:

```
df_merged = df_merged[df_merged["PROD_QTY"] != 200]
```

2.3 Feature Engineering — Brand & Pack Size

Two new features were extracted from PROD_NAME:

```
data["brand"] = data["PROD_NAME"].str.split().str[0]
```

```
data["packet_size"] = data["PROD_NAME"].str.extract(r"(\d+)g", flags=re.IGNORECASE)
```

Brand names were then standardised — 8 abbreviations/variants were mapped to their correct names (e.g. RRD → Red, NCC → Natural, GrnWves → Grain), resulting in **21 clean unique brands**.

2.4 Product Validation — Removing Non-Chip Products

All product names were split into individual words and counted. After removing digits and special characters, the word 'Salsa' appeared suspiciously. Further inspection revealed 5 products that were dips (sauces), not chips. These were removed:

- Old El Paso Salsa Dip Tomato Med 300g
- Old El Paso Salsa Dip Tomato Mild 300g
- Old El Paso Salsa Dip Chnky Tom Ht300g
- Woolworths Medium Salsa 300g
- Woolworths Mild Salsa 300g

After removal, all remaining product words were verified as chip-related (brand names, flavors, chip styles) — confirming the dataset is clean.

3. Customer Segment Analysis

3.1 Total Sales by Segment

Top 3 segments by total chip sales:

Rank	Segment	Total Sales	Driver
#1	Older Families + Budget	\$168,363	High frequency + family size
#2	Young Singles/Couples + Mainstream	\$157,622	Largest customer base
#3	Retirees + Mainstream	\$155,677	Large & consistent customer base

Key observation: Premium customers consistently spend LESS on chips across all life stages, confirming chips are perceived as a Budget/Mainstream product.

3.2 What Drives Spending?

Three metrics were examined to understand what truly drives total sales for each top segment:

Segment	Customers	Avg Spend/Txn	Frequency/Year	Primary Driver
Older Families + Budget	4,675	\$7.27	4.95x	Frequency + Volume
Young Singles + Mainstream	8,088	\$7.56	2.58x	Customer Volume
Retirees + Mainstream	6,479	\$7.25	3.31x	Customer Volume

3.3 Average Price per Unit — Statistical Testing

To determine if Mainstream Young/Midage Singles spend significantly more per unit than other tiers, an independent t-test was performed:

Comparison	t-statistic	p-value	Conclusion
Mainstream vs Premium	30.09	0.0000	Significant — Mainstream pays more
Mainstream vs Budget	33.35	0.0000	Significant — Mainstream pays more
Budget vs Premium	-2.70	0.0070	Significant — Budget pays slightly less
Mainstream vs Budget+Premium	38.77	0.0000	Significant — Mainstream pays more

All differences are statistically significant ($p < 0.05$). Mainstream Young/Midage Singles willingly pay more per chip unit — making them the highest value target segment.

4. Brand & Pack Size Preferences

4.1 Brand Affinity — Mainstream Young Singles/Couples

Affinity analysis measures how much a segment over- or under-indexes on a brand versus the total population. Affinity > 1 means the segment buys this brand more than average.

Brand	Affinity	Interpretation
Tyrrells	1.22	Buys 22% MORE than average — strongly preferred
Twisties	1.21	Buys 21% MORE than average — strongly preferred
Tostitos	1.19	Buys 19% MORE than average — preferred
Kettle	1.18	Buys 18% MORE than average — preferred
Pringles	1.17	Buys 17% MORE than average — preferred
Woolworths	0.51	Buys 49% LESS than average — strongly avoided
Sunbites	0.54	Buys 46% LESS than average — avoided

4.2 Pack Size Affinity — Mainstream Young Singles/Couples

Pack Size	Affinity	Interpretation
270g	1.25	Buys 25% MORE — preferred (party size)
380g	1.24	Buys 24% MORE — preferred (party size)
330g	1.21	Buys 21% MORE — preferred (party size)
110g	1.16	Buys 16% MORE — preferred (on-the-go)
134g	1.17	Buys 17% MORE — preferred (on-the-go)
175g	0.96	Slightly below average — standard size avoided
160g–200g	0.51–0.61	Strongly avoided — medium sizes unwanted

5. Strategic Recommendations for Julia

Based on the full analysis, the following data-driven recommendations are presented to support the upcoming category review.

PRIORITY 1 — Protect & Grow the Top 3 Segments

Segment	Action
Older Families + Budget	Ensure strong availability of value-sized packs (175g). Promote Kettle brand deals. These customers are the most price-sensitive.
Young Singles/Couples + Mainstream	Place Kettle, Tyrrells, Twisties, Tostitos, Pringles prominently. Offer large party packs (270g–380g) and small snack packs (110g–150g).
Retirees + Mainstream	Consistent and reliable shoppers. Focus on Kettle and Smiths. Standard 175g packs are preferred.

PRIORITY 2 — Shelving & Ranging Strategy

Decision	Recommendation
Increase shelf space	Kettle — universally preferred across ALL segments. It commands a higher price (~\$9.20–\$9.60) and is the most profitable brand.
Feature prominently	Tyrrells, Twisties, Tostitos, Pringles — especially in aisles frequented by young shoppers.
Reduce shelf space	Woolworths generic brand and Sunbites — consistently under-indexed in the most valuable segments.
Pack size focus	Prioritise 175g (universal favourite) + large party packs (270g–380g) + small snack packs (110g–150g).

PRIORITY 3 — Pricing & Promotions

Insight	Action
Mainstream pays the most per unit	Target Mainstream Young/Midage Singles with mid-to-premium priced products. They are statistically the most profitable segment.
Budget families are high-frequency buyers	Run buy-one-get-one-free promotions (buy 2 get 1 free) for Budget Older/Young Families — they already shop frequently.
December is peak month	Increase stock of party-size packs and trending brands (Kettle, Pringles, Doritos) in November–January.
February is lowest month	Consider a targeted promotion in February to lift the seasonal dip — especially for Mainstream Young/Midage Singles.

Bottom Line for Julia: The chip category is healthy and stable across all 12 months. The three highest-value segments — Older Families (Budget), Young Singles/Couples (Mainstream), and Retirees (Mainstream) — together account for the majority of revenue. Kettle is the undisputed category leader. Mainstream Young and Midage Singles/Couples are the most price-tolerant segment and should be the primary target for premium product placement and marketing investment.